Churn Prediction & Recommendation System

with Uplift Modeling — A Causal Inference Approach to Customer Retention

CAI 6826 SPRING 2026 | UNIVERSITY OF FLORIDA | ARYAN ATRE

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The Problem with Traditional Churn Models

❌ The Old Question

"Who will churn?"

Traditional models identify high-risk customers — but ignore whether they'll actually *respond* to intervention. Result: **30–40% of retention budgets** wasted on "Lost Causes" who leave regardless.

✅ The Better Question

"Who will respond if we act?"

Uplift Modeling estimates the **individual treatment effect**:

$$\tau(x) = P(Y=1 \mid T=1, X) - P(Y=1 \mid T=0, X)$$

This isolates **"Persuadables"** — customers who churn without intervention but stay when targeted.

Compare Meta-Algorithms

T-Learner, S-Learner, X-Learner head-to-head

SHAP Explainability

Transparent, interpretable individual predictions

Production System

End-to-end recommendation dashboard

Measurable ROI

Validated business impact with projected 1,204% return

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System Architecture & Key Components

Two-Stage Modeling Pipeline

Stage 1 – Churn Prediction

Random Forest (AUC 0.82) · LightGBM (AUC 0.84) · **XGBoost (AUC 0.84) ✓ Selected** Output: P(Churn | X)

Stage 2 – Uplift Modeling

T-Learner (Qini −0.71) · **S-Learner (Qini 0.33) ✓ Selected** · X-Learner (Qini −1.90) Output: Treatment Effect τ(x)

Deployment & Business Output

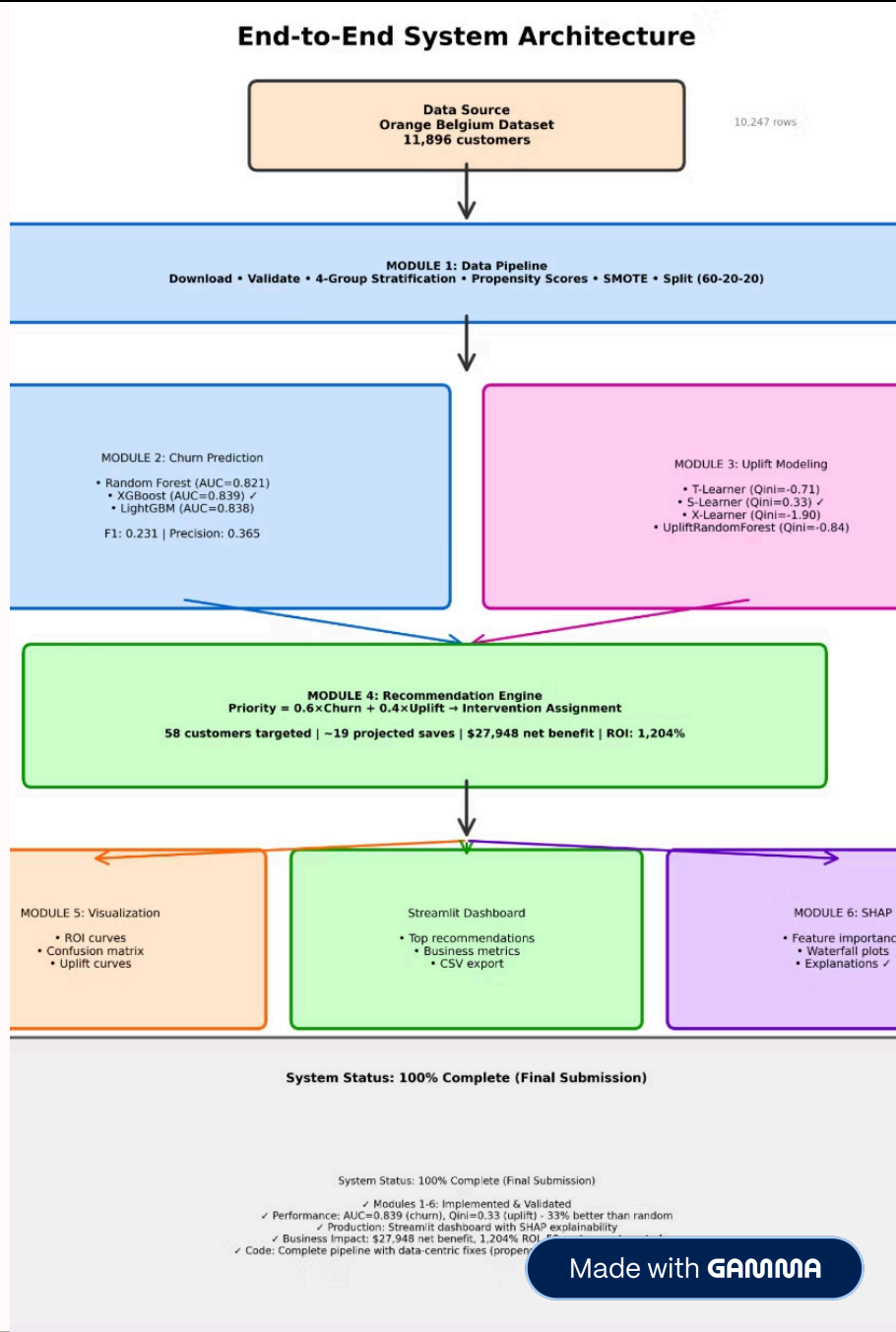
Data: **Orange Belgium — 11,896 customers** · 4-group stratification · propensity scores · within-arm SMOTE

Streamlit Dashboard

ROI waterfall · customer recs · SHAP explanations

$27,948 Net Benefit

58 customers targeted · 1,204% projected ROI



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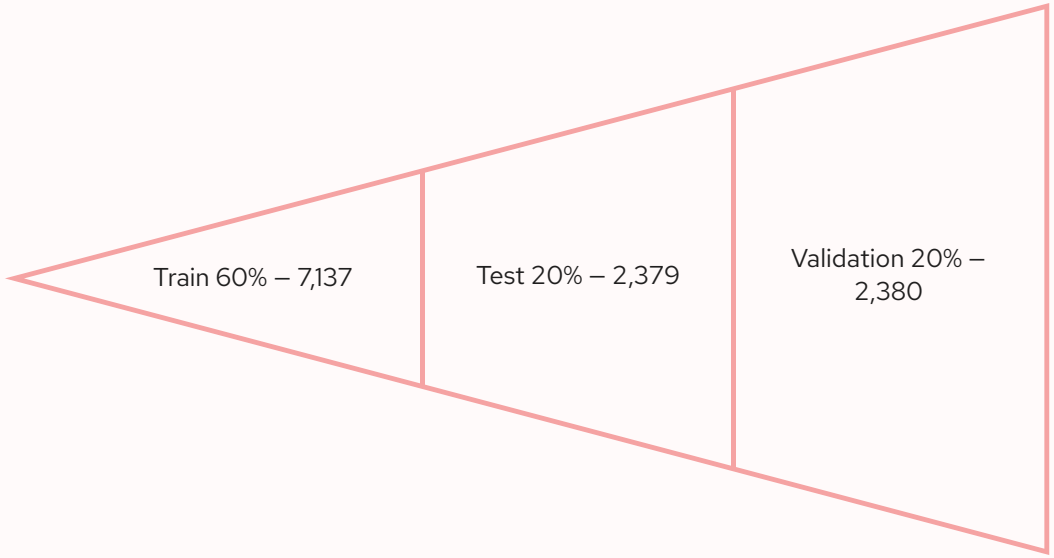
Dataset & Implementation Pipeline

Dataset at a Glance

Key Techniques Applied

- **Propensity Score IPW:** Corrects 75/25 imbalance → ATE −0.0029 → +0.0014
- **Within-Arm SMOTE:** Boosts positive rate 3.4% → 15% (4× more signal)
- **Class-Weight Balancing:** scale_pos_weight=28.2 for XGBoost



4-group stratified splits ensure every rare combination of treatment × churn status is represented in train, validation, and test sets — preserving causal signal throughout.

11,896 Customers · 178 Features

Churn rate 3.43% (408 churners) · Treatment 75.7% / Control 24.3% · **28:1 class imbalance**

⚠️ Core Challenges

Outcome sparsity · treatment imbalance · fully anonymized features

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Final Results & Evaluation Metrics

0.84

XGBoost AUC

Best churn prediction model, F1 score 0.231

0.33

S-Learner Qini

33% improvement over random targeting

58

Customers Targeted

High-uplift persuadables identified from 11,896

$27,948

Net Benefit

After $2,320 intervention cost · ~19 projected saves

1,204%

Projected ROI

Validated through Qini curve analysis

Metric	Value	Interpretation
Best Churn Model	XGBoost	AUC 0.839, F1 0.231
Best Uplift Model	S-Learner	Qini 0.33 — 33% above random
Net Benefit	$27,948	After $2,320 intervention cost
ROI	1,204%	Projected impact on test set

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Key Challenges & How They Were Solved

1

The Zero Qini Mystery

All uplift models returned Qini = 0.0000 for **2 weeks**. Systematic debugging uncovered 6 root causes: CausalML API bug, string vs. int treatment labels, wrong base class, dataset misalignment, data leakage, and missing propensity integration.

2

Extreme Data Imbalance

3.4% churn + 75/25 treatment split = dangerously sparse segments. Resolved with propensity score IPW + within-arm SMOTE + 4-group stratification — revealing a **hidden ATE: −0.0029 → +0.0014**.

3

Theory vs. Reality

Textbooks favor T-Learner, but S-Learner won. Why? T-Learner trains on 170/60 samples (high variance); S-Learner uses 230 combined (lower variance). **Threshold: ~500 samples/group for T-Learner to win.** Empirical validation beats theory.

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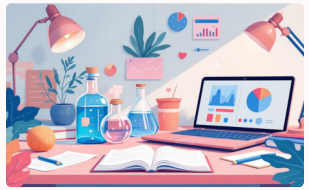
Lessons Learned: Data Science in Practice

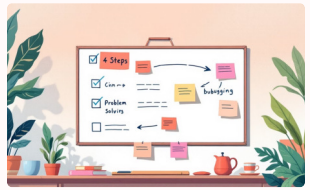
💡 Data Quality Trumps Algorithm Choice

The Qini jump from 0.0 → 0.33 came entirely from **data fixes**, not model changes. Propensity weighting + stratification outperformed hyperparameter tuning. *Andrew Ng's data-centric AI in action: fix your data first.*



📚 Theory Guides, Doesn't Govern

T-Learner's theoretical edge holds — but only past ~500 samples/group. Our 170/60 split was below threshold. **Always validate theoretical claims empirically on your specific dataset.**



🔧 Systematic Debugging is a Meta-Skill

Hypothesize → Isolate → Test → Validate. This structured loop — not random trial-and-error — resolves model failures, pipeline errors, and production incidents alike. A transferable skill that compounds over time.

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Impact & Significance

🎓 Academic Contribution

- Causal inference under extreme imbalance
- Extends meta-learner comparison literature
- Documents bias-variance tradeoffs with sparse data
- Reproducible debugging methodology for uplift models

🛠️ Technical Contribution

- Production-ready uplift modeling template
- Reusable pipeline: preprocessing → modeling → explainability → deployment
- Modular architecture adaptable to any domain

💰 Business Impact at Scale

For 100K customers: targeting **15K persuadables** vs. 30K high-risk saves an estimated **$600K annually** — by eliminating spend on non-responders.

🌐 Broader Applications

- **E-commerce:** Cart abandonment recovery
- **SaaS:** Upgrade campaign targeting
- **Healthcare:** Treatment response prediction
- **Politics:** Voter mobilization optimization

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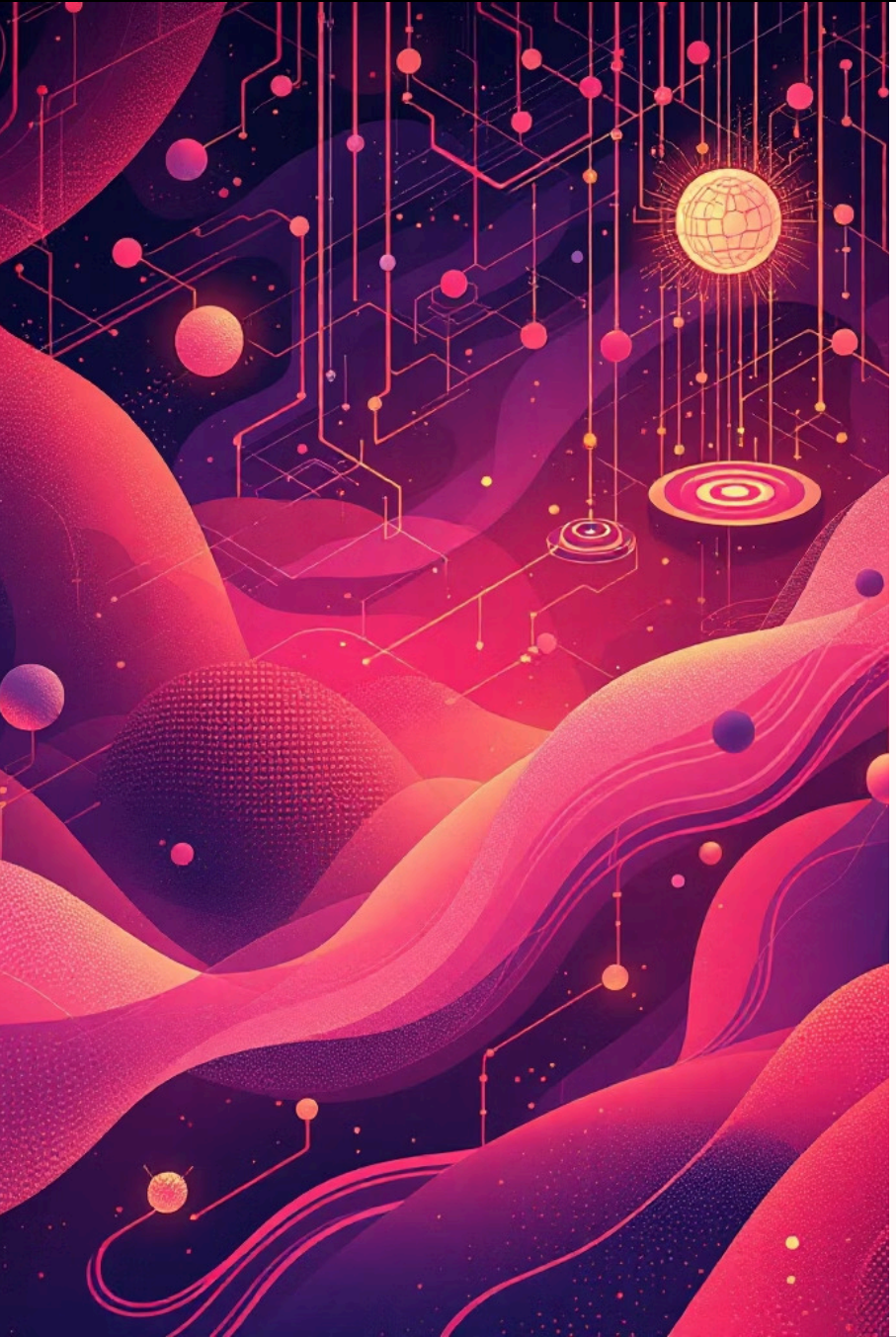

Future Improvements & Extensions

Deep Learning Uplift

DragonNet & TARNet architectures capture complex treatment-feature interactions. Potential: Qini 0.33 → **0.45+** on larger datasets.

Multi-Armed Uplift

Model multiple interventions simultaneously — Discount vs. Loyalty vs. Premium Support — matching real business trade-off scenarios.

Temporal Modeling

Survival analysis answers *when* churn occurs and how intervention *timing* shifts outcomes. Enables dynamic, personalized strategies.

Reinforcement Learning

Sequential interventions over time: Discount at M6 → if fails, upgrade at M9. The frontier of causal ML in production.

⚡ These extensions require richer longitudinal datasets with repeated customer interaction records over time.

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Conclusion

What Was Built & Proven

01

End-to-end uplift modeling system on real telecom data (Orange Belgium, 11,896 customers)

02

Debugged 6 critical pipeline failures — Qini 0.0 → **0.33**

03

Applied propensity scores + within-arm SMOTE + 4-group stratification

04

Deployed production-ready Streamlit dashboard with SHAP explainability

05

Demonstrated **1,204% projected ROI** and $27,948 net benefit

The Core Takeaway

> Real data science means navigating constraints, validating assumptions, and making pragmatic choices — not just implementing algorithms.

Shifting the question from **"Who will churn?"** to **"Who will respond?"** is the key to maximizing retention ROI at scale.

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